

Day 7 – Cyber Security & Hacking

Building Legal Hacking Labs (Ethical Hacking Foundations)

Concept: Why Legal Labs Matter

Hacking **without permission** is:

- Illegal
- Punishable under cyber laws (IT Act, CFAA, etc.)
- Career-ending

Ethical hackers **never test random websites**.

They **reproduce real-world systems** inside **isolated environments** where:

- You own the systems
- You control the network
- You have written permission

This transforms hacking into **skill development**, not crime.

Definition: What is a Legal Hacking Lab?

A **legal hacking lab** is a **self-owned, isolated environment** designed to intentionally contain vulnerabilities so security professionals can practice attacks safely and lawfully.

Technical Definition

A hacking lab is a **sandboxed infrastructure** consisting of:

- Vulnerable operating systems
- Insecure web applications
- Simulated networks
- Controlled attacker machines

All activity occurs **offline or in private networks**.



Core Components of a Hacking Lab

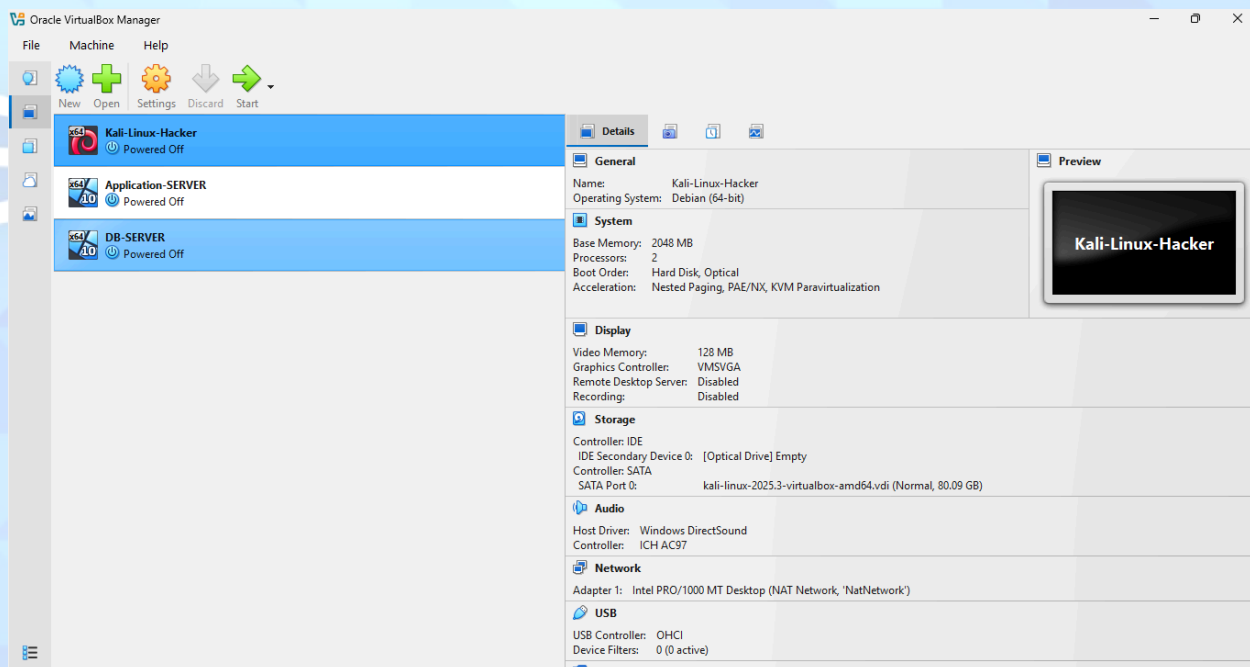
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1 Virtualization Layer

Allows multiple systems to run on one machine.

Examples:

- VirtualBox
- VMware



2 Attacker Machine

This simulates the hacker.

Common choice:

- Kali Linux

Contains: Nmap, Burp Suite, Metasploit, SQLmap

3 Vulnerable Targets

These systems are **intentionally insecure**.

Examples:

- DVWA (Damn Vulnerable Web App)
- PortSwigger
- OWASP Juice Shop

Day 7 Practical Example (Legal Setup)

Lab Architecture

[Kali Linux] ----> [DVWA Web App]

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Kali-Linux-Hacker [Running] - Oracle VirtualBox
File Machine View Input Devices Help

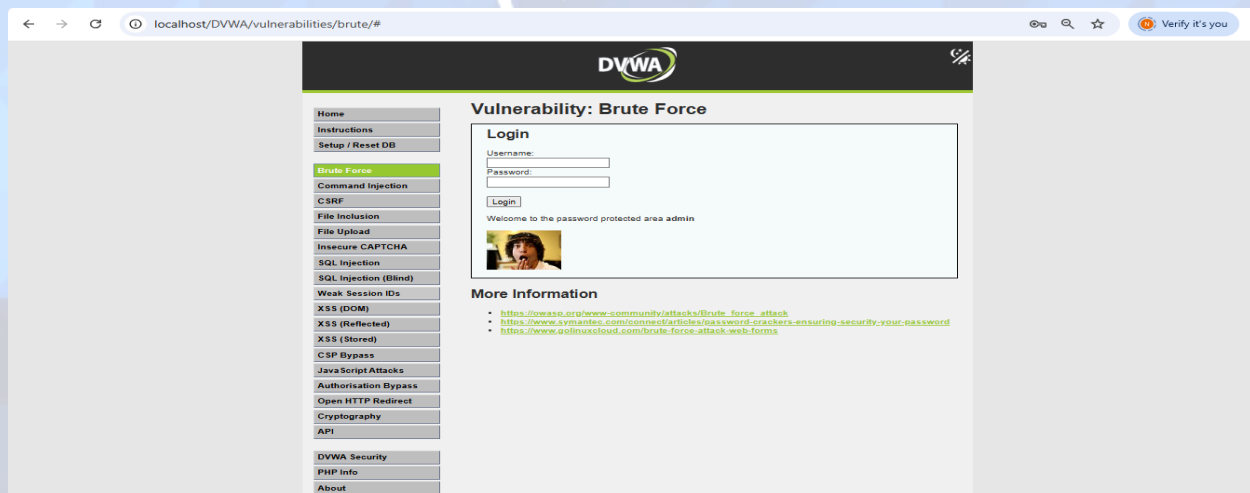
(kali@kali)-[~]
$ nmap 192.168.0.100
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-04 10:57 EST
Nmap scan report for 192.168.0.100
Host is up (0.0028s latency).
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
443/tcp   open  https
445/tcp   open  microsoft-ds
902/tcp   open  iss-realsure
912/tcp   open  apex-mesh
3306/tcp  open  mysql
16992/tcp open  amt-soap-http

Nmap done: 1 IP address (1 host up) scanned in 4.65 seconds

(kali@kali)-[~]
$

```

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Sample Lab Exercise (Safe & Legal)

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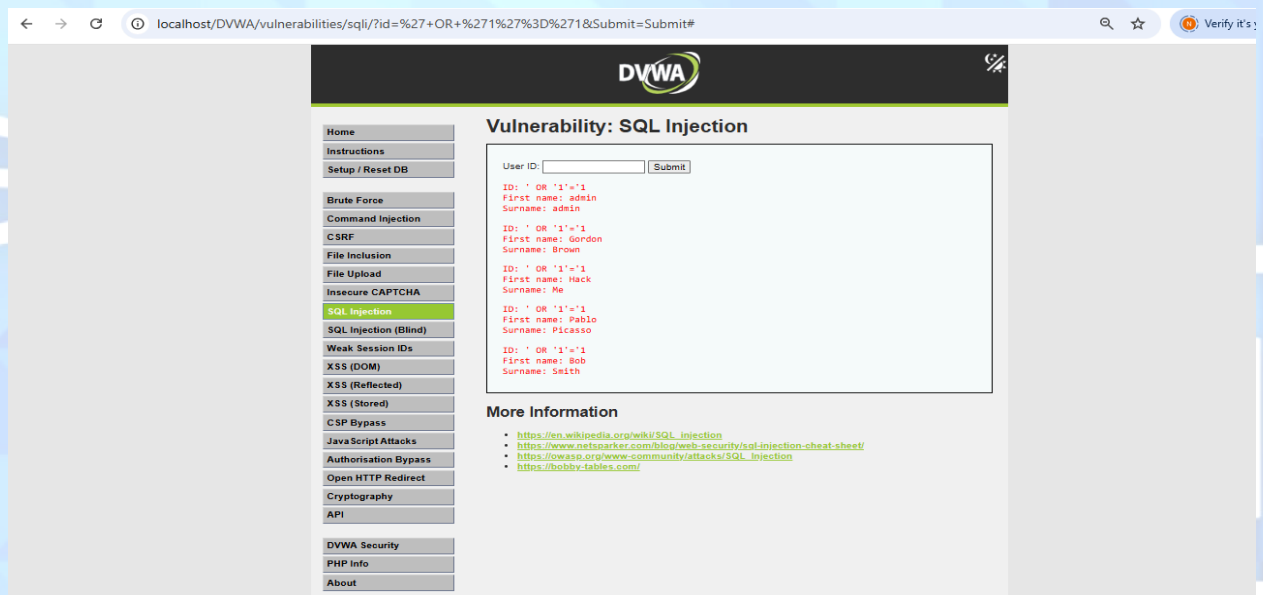
SQL Injection (Authentication Bypass)

Target

DVWA Login Page

Input (Test Payload) : ' OR '1'='1

Output



PortSwigger-Labs

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